4.3 Construction of representative group

In the next step, a scheme with a conditional objective is implemented. The objective is to dampen high numerical variations of parameters describing events with the condition of retaining the identity of individual events. Therefore, the most important task is to hit the right level of homogeneity as a whole and independence of each case. It is necessary to establish an equal base from the starting point if we are to satisfy the homogeneity aspect. Therefore, we initiate the transformation from a circle with area equal to one ($S = 1, r = \sqrt{\frac{1}{\pi}}$) resembling a typical probability density function which is nonnegative everywhere and its integral over the entire space is equal to one. The transformative procedure for input set (i.e. Biomass properties and operating condition) is as follows:

The total area is divided into 4 sections, on the account of presence of 4 entities in the input set (i.e. biomass as a whole plus the three independent operating condition parameters, temperature (T), equivalent air ratio (ER), and steam to biomass ratio (SBR)). That said, allocation of a section to an entity is not exclusive for all, and distribution of area shares is done with following considerations:

1. The extent of involvement of a specific parameter to constitute an area is commonsurate with its significance based on the subject knowledge of the process. For example, temperature, in addition to having its own designated section, plays an auxiliary role in other sections.
2. The distribution is set up to possess the attribute of true randomness. The aim of increasing randomness intentionally, and at the earliest step of the process is to produce immunity for the model against high levels of uncertainty.

3. A lower and upper bound in calculation of each section share is considered to prevent dominance of any section. In our case, elimination of SBR section is avoided for cases where SBR = 0 to keep from alternating between 3 and 4 sections.
4. An internal hierarchy among sections is devised in terms of inclusiveness (Fig. 41).

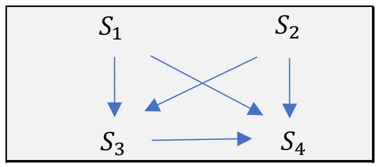


Figure 41. $S_1, S_2, S_3, and\ S_4$ are influenced respectively by [only SBR], [T or T & ER], [Biomass, T, SBR or ALL], and [ALL]

Based on the scale of our numerical values for the input set [d, k, T, ER, SBR], the following formulation and arrangement seemed to meet our conditions.

$$s_1 = \frac{3 * SBR}{100} + 0.05 \qquad (55)$$

$$s_2 = 0.7 * max\left\{\frac{T-1000}{T}, \frac{T-1000}{T} + ER\right\} \qquad (56)$$

$$s_3 = \left(\frac{1+d^2}{1+d^2+\frac{10^3 k^2}{T}}\right) * (1-(s_1+s_2)) \qquad (57)$$

$$s_4 = 1-(s_1+s_2+s_3) \qquad (58)$$

Transformation of a typical set of [d, k, ER, SBR, 1000/T] to $[S_1, S_2, S_3, S_4]$ (Figure 42).

d	k	ER	SBR	1000/T
1.54	2.12	0.22	2.70	1.43

Figure 42. Illustration of the example $[s_1 = 0.13, s_2 = 0.45, s_3 = 0.22, s_4 = 0.20]$

The representation illustrated in Figure 30 has several advantages as discussed, however in following our strategy to enforce an effective system of checks and balances preventing the unfairly favoring certain subdivisions, it is decided to make use of the wellknown structure of a bicameral legislature system in which each state is represented both based on “population or size” and also “regardless of the size”. Here, we retain the property of size (i.e. area of circular sector) and for the second basis we assign “exactly one full quadrant” to each section regardless of their size (Fig. 43). Geometrically, the similarity of each partition and perpendicularity of lines separating the four areas are desirable features.

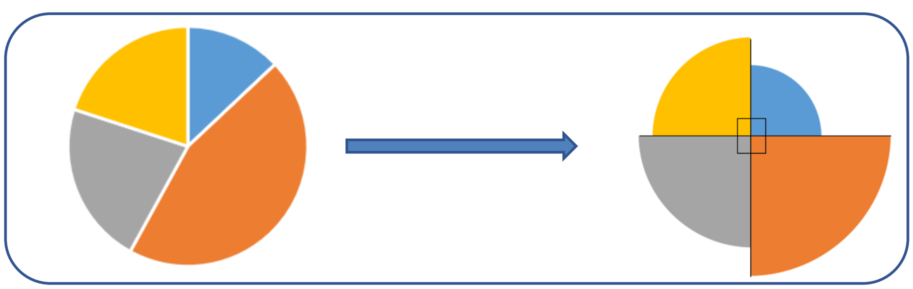
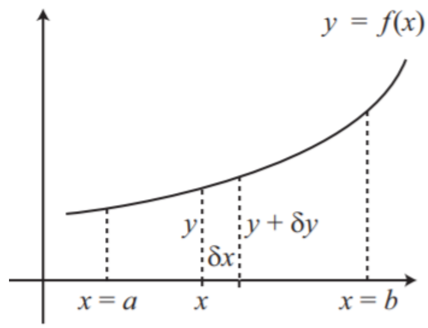


Figure 43. Assign “exactly one full quadrant”

Then, the revolutionary solid volume of quadrants is calculated to turn the 2D information to 3D to attain full geometric representation in terms of "number of dimensions". In general we have:



$$V = \sum_{x=a}^{x=b} \partial V = \int_a^b \pi y^2 dx \qquad (59)$$

Sum of the volumes for each quadrant with respect to the x and y axis yields three quarters of an sphere with the same radius.

Therefore:

$for\ i = 1, 2, 3, 4\ \ r_i = 20 * \sqrt{\frac{S_i}{\pi}}$ (multiplied by 20 only for scaling numerical values), and

$v_i = \pi r_i^3$,

Another advantage of the transformation from slices with different angles to full quadrants is that quadrants can be divided into a straight segment (i.e. an isosceles right triangle) and a curvilinear segment (i.e. a cord with a fixed arc length) only by addition of one line (this is viewed as efficient binary classification).

Since the role of curvilinear segment is naturally have been addressed via the volumetric variable (v_i), we turn our attention to the straight segment which exhibited a strong relationship with its counterpart from output set (i.e. the product gas composition). The product of ratio of altitudes ($l_i = \frac{\sqrt{2}}{2} r_i$) in right triangles (Fig. 44) showed 72%

correlation with the same variable built from output set. This strong correlation implies the efficacy of the steps taken up to this point in building highly informative characteristics.

Figure 44. Altitudes and binary areas of the kite

Additional averaging operations were done.

Sum of the equivalent radius $R_j = \sum_1^4 \quad r_i,$ (60)

Sum of the areas made out of binary altitudes $S_j = \frac{l_1 l_2}{2} + \frac{l_2 l_3}{2} + \frac{l_3 l_4}{2} + \frac{l_4 l_1}{2}$ (61)

A weighted sum of volumes: $V_j = min\,(273 + \frac{v_1}{16} + \frac{v_2}{28} + \frac{v_3}{2} + \frac{v_4}{44}, 940)$ (62)

It can be seen that the selected weights are actually the molecular weights of the product gas composition. This type of assignment was implemented based on the assumption that we would gain more through this great leap of increasing similarity between two groups of input and output than what we may lose due to lack of an inherently one to one relation between these volumes and gas components. An anology to this situation is where a 1v1 defending tactic is used in soccer alongside the overall strategy of defending as a group.

Ratio of diagonals of the kite: $L_j = \frac{l_1}{l_4}$ (63)

Composite variable: $N_j = \frac{1}{2}(\frac{L_j * V_j}{T_j S_j * lnln\,(R_{j\_max} - R_j)} + 12) * \frac{2.3 * R}{F}$, (64)

where T_j is the temperature in ℃, R=the universal gas constant, and F is the Faraday constant.

The rationale for involvement of Faraday constant is the following. To make the mathematical object created in the previous process “electrically charged” with the aim of enabling or possibly enhancing its frequency response when tested with respect to the variable of “computational time”.

In the last step of “representative” preparation we aim to recuperate its overall “uniformity” which may have been weakend due to multiplexity of steps taken after the equity circle (S=1).

For this purpose, the structure of Fibbonacci sequence is utilized which has a prominent feature that $\frac{f_{n+1}}{f_n} = \varphi$, where φ is the golden ratio ($\frac{1+\sqrt{5}}{2}$).

The integers in Fibbonacci numbers are constructed as following, starting with $f_1 \& f_2 \& f_3$ as seed values, and then building upon them by the convention that nth Fibonacci number is the sum of the previous two Fibonacci numbers:

$$f_1 = N_j + e^{-N_j}, f_2 = f_1 + e^{-N_j}, f_3 = e^{-N_j} * f_2\ , f_4 = f_3 + f_2, f_5 = f_4 + f_3, f_6 = f_5 + f_4,\ f_7 = f_6 + f_5, f_8 = f_7 + f_6, f_9 = f_8 + f_7, f_{10} = f_9 + f_8, f_{11} = f_{10} + f_9 \quad (65)$$

The golden ratio (approximately 1.6) starts to emerge after f_6 based on our computations and gets closer to actual value of ($\frac{1+\sqrt{5}}{2}$) moving forward. Therefore, we selected the set of $F_j = [f_6, f_7, f_8, f_9, f_{10}, f_{11}]$ as our final representatives that will go through the functional analyses. (It should be noted for convenience the subscripts 1 to 5 is used instead of 6 to 11 from now on) Figure 45 summarizes the major steps taken to obtain what we now percieve as a cohesive representation.

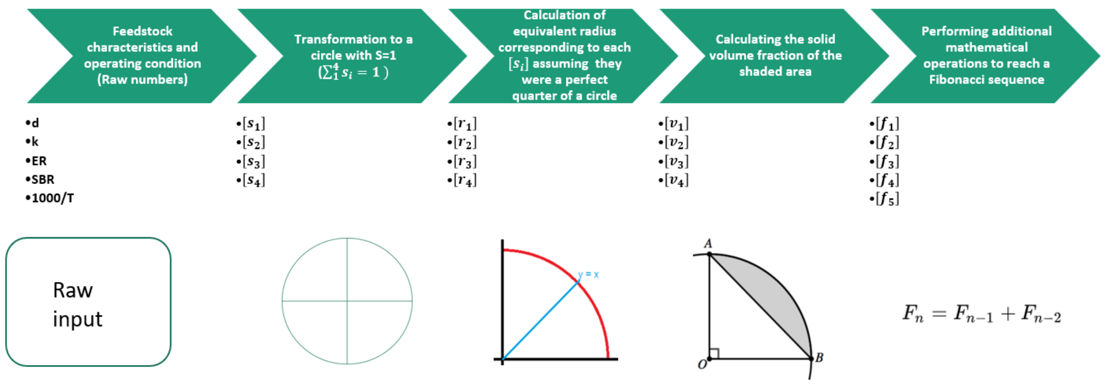


Figure 45. The progress of representative preparation

The following tables show few examples of final sets built from original data (Figure 46):

f1	f2	f3	f4	f5
16	26	42	69	111
15	24	40	64	104
15	24	39	62	101
15	24	39	62	101
15	25	40	64	104
16	26	42	69	111
16	26	42	69	111
16	26	43	69	112
16	27	43	70	113
17	27	43	70	114
20	32	53	85	137
17	27	44	71	114
15	25	40	65	106

: Fibonacci sequence obtained from Given information of the problem (Feed& Operating condition)

Figure 46. Examples of Fibonacci sequences constructed from composite variable Nj. (Page 1 of 2)

f1	f2	f3	f4	f5
18	29	47	76	123
18	30	48	77	125
21	35	56	90	146
23	37	59	96	155
27	43	70	114	184
21	34	55	89	144
22	36	58	93	151
24	39	63	102	164
23	37	60	96	156
22	36	58	93	151
30	49	80	129	208
23	37	59	96	155
22	35	57	91	148

: Fibonacci sequence obtained (using similar procedure) from synthesis gas composition (H2, CO, CO2 and CH4)

34	55	89	145	234
33	54	87	141	229
36	58	94	153	247
37	60	98	158	256
42	68	110	178	288
37	60	97	158	255
38	62	100	162	262
40	65	105	171	276
39	63	103	166	269
39	62	101	163	265
50	82	132	214	346
39	64	103	166	269
37	60	97	157	254

+

Figure 46. Examples of Fibonacci sequences constructed from composite variable Nj. (Page 2 of 2)

The following plots illustrate how the distance between center of A and B differs among the 200 cases under study (First plot: a typical example, Second plot: an abnormal instance). The distance between two centers is of critical importance (Fig. 47).

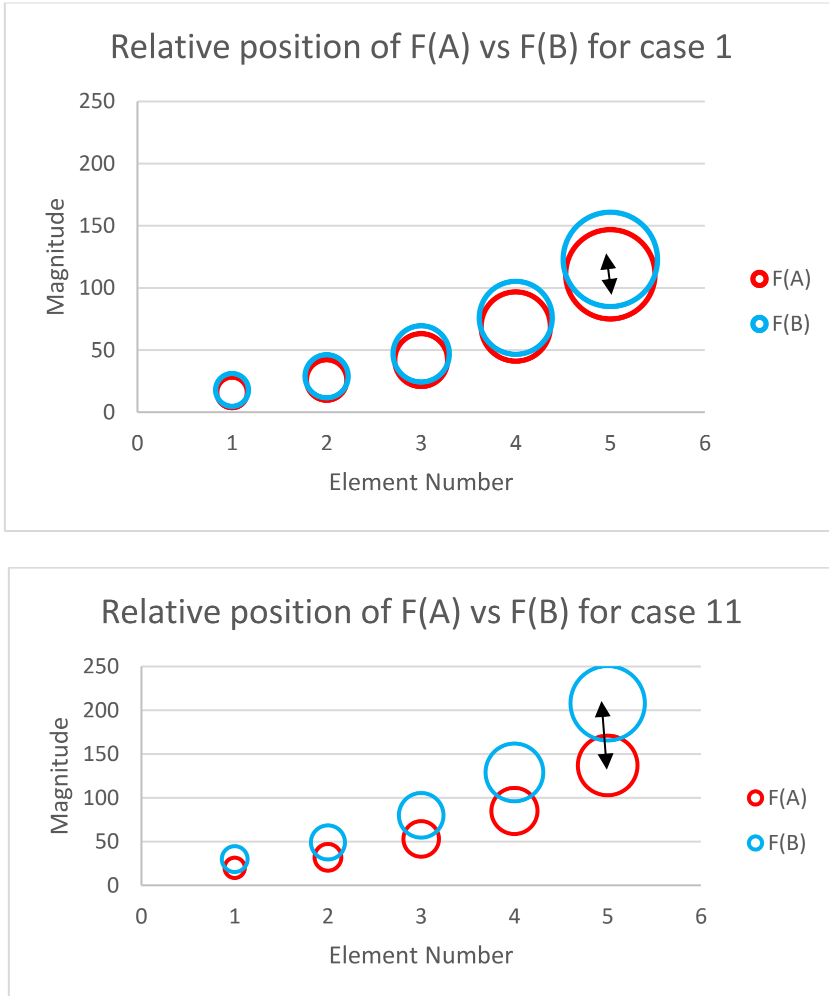


Figure 47. The centric differences of two associated Fibonacci sequences, up: typical, bottom: abnormal

At this juncture, we have successfully converted our original problem into harmonious integer sequences, representing the event (i.e. feedstock, operating condition, and product gas composition). It should be noted that carbon conversion efficiency (CCE%) and product gas yield (Nm3/kg) are the output variables that should be predicted by the model, and not included as part of the event embedded in the sequences. That will keep us two degrees of freedom away from "totality" of the process. The overall transformations practically meet the specifications of a formal mathematic "Group", defined in Group theory: “In mathematics, a group is an algebraic structure consisting of a set of elements equipped with an operation that combines any two elements to form a third element and that satisfies four conditions called the group axioms, namely closure, associativity, identity and invertibility. One of the most familiar examples of a group is the set of integers together with the addition operation, but the abstract formalization of the group axioms, detached as it is from the concrete nature of any particular group and its operation, applies much more widely. It allows entities with highly diverse mathematical origins in abstract algebra and beyond to be handled in a flexible way while retaining their essential structural aspects. The ubiquity of groups in numerous areas within and outside mathematics makes them a central organizing principle of contemporary mathematics.”[82]

The next step is to design a mathematical set up to serve us as a computational labrotary. Then by performing specific analyses which will be elaborated shortly, we will predict the most probable outcome of biomass gasification in fluidized bed reactor for a given input set.

The central idea of our statistical approach is “indirectness”, therefore basing the inference about the identity of an inquiry on some type of performance along an arbitrary path. The inclusiveness of the designed path and stages upon which the performance is to be monitored has direct effect on the quality and amount of information we gain. In our study, a closed, multilateral path is laid out starting from Fibbonacci sequences, followed by series of matrix operation, and ultimately reaching a fully symmetric destination (similar to the invariant point in geometric projections) that reveals accurate and deep understanding about the original problem.

4.4 Design of neural based computational lab

Here, we first give a brief and general introduction on neural networks, and then explain what role they play in our modeling.

Neural networks are conceptually quite simple computing systems, inspired by structure of biological neurons in human brain. Neural networks use computational methods to "learn" information from data without relying on a predetermined equation as a model. Deep learning is especially suited for image recognition, which is important for solving problems such as facial recognition, motion detection. "Such systems progressively learn through examples without task specific programming. Various architectures of ANNs are being used of which Multilayer Perceptron (MLP) is the most popular one. MLP consists of an input layer, a hidden layer and an output layer of neurons. While the neurons in the input layer are typically linear, the neurons in the hidden layer have nonlinear activation functions. The neurons in the output layer may be linear or nonlinear. The input layer consists of input neurons with associated weight. The hidden layer receives the signals from the input layer. There's also a threshold parameter known as bias which is associated with each neuron in the hidden and output layers" [83]. "An elementary neuron with *R* inputs is shown below. Each input is weighted with an appropriate *w*. The sum of the weighted inputs and the bias forms the input to the transfer function *f*. (Fig. 48)*"* [84]

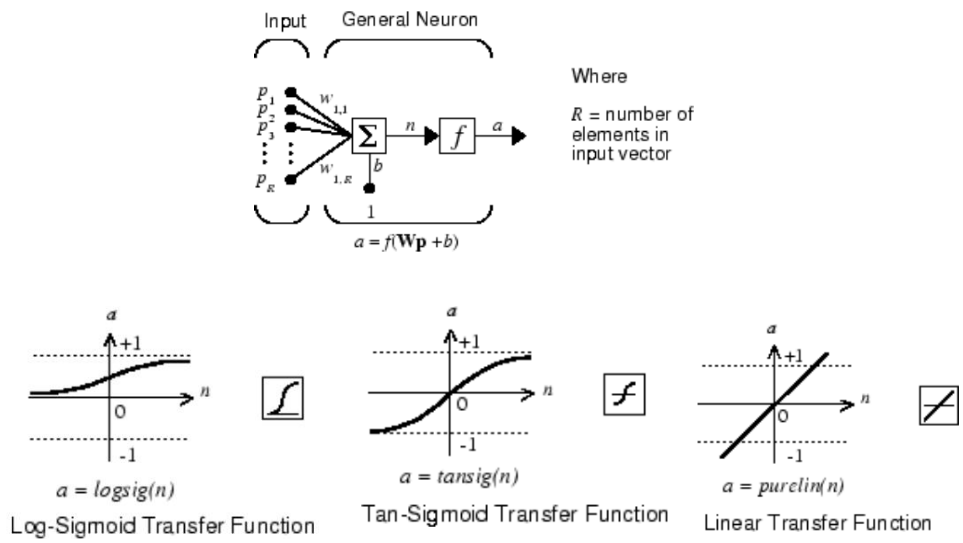
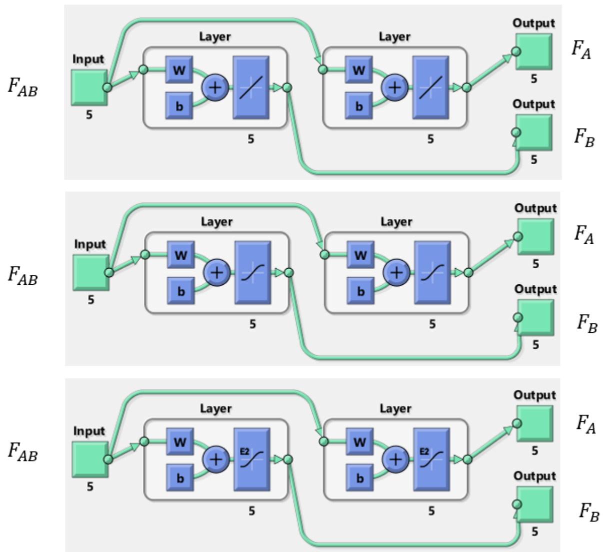


Figure 48. The structure of a general neuron as foundational unit of neural networks [84]

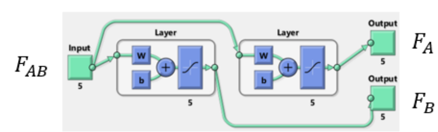
Development of three stable neural network unit featuring vanished gradients to serve as a computational laboratory:

Three neural network unit with three different transfer functions (i.e. linear, hyperbolic tangent sigmoid, and elliot symmetric sigmoid) were trained by the sets of Fibbonacci sequences with the configurations illustrated in Figure 49. (The ABT is chosen to exemplify the NN details).



- All NN's are composed of an input layer that receives F_{AB}, 2 hidden layers each with 5 neuron designated to predict F_A and F_B separately.
- **ABL, ABT, ABE** are 3 NN's trained based on **L**inear, **T**angential and **E**lliptical transfer function, respectively.

- There are **60** elements in ABT.
- **50** elements in terms of weight (w) (i.e. one matrix of $[\,]_{5\times5}$ for each layer).
- And, **10** elements in terms of bias (b) (i.e. one vector of $[\,]_{5\times1}$ for each layer).
- For example, the following numerical configuration, is resulted upon training the "ABT" neural network using Levenberg-Marquardt backpropagation algorithm with aim of minimizing sum of squared errors in predicting F_A and F_B.



-1.42	2.51	1.77	1.28	1.35	-0.81
-0.61	0.48	-3.08	0.94	-1.98	3.09
-2.01	-2.19	-2.27	-0.24	0.93	2.89
-2.20	-1.59	-1.98	0.28	-1.89	2.72
3.13	-1.04	1.06	-1.44	-0.93	1.55

The first layer in the figure connected to output F_B

1.43	-0.15	0.12	3.54	0.57	-4.69
2.42	-0.55	2.75	0.28	1.06	-3.94
-2.98	-0.38	2.10	-1.06	0.62	0.85
2.30	-1.04	0.77	2.35	-1.56	-0.44
2.04	0.06	-0.90	2.79	-1.47	0.67

The second layer in the figure connected to output F_A

```
dimensions:

        numInputs: 1
        numLayers: 2
       numOutputs: 2
   numInputDelays: 0
   numLayerDelays: 0
numFeedbackDelays: 0
numWeightElements: 60
       sampleTime: 1

connections:

      biasConnect: [1; 1]
     inputConnect: [1; 1]
     layerConnect: [0 0; 0 0]
    outputConnect: [1 1]
```

Figure 49. The detailed configuration of ABT (i.e. the neural network set up using hyperbolic tangent as transfer function)

Following the training, the neural network units shows properties of a bounded function with respect to the inputs (i.e. F_{AB}), meaning for all the inpust they return identical vectors of :

$$F_A = [1,1,1,1,1];\ F_B = [1,1,1,1,1];$$

This seemingly short circuited configuration provides critical information on the hollistic state of data distributions. Specifically, the weights and biases calculated upon the training indirectly capture the distribution of input sets with respect to their corresponding output set. This can be viewed as an unconventional method of averaging distributions.

We will show how this particular quality can form a structure preserving map, and yield a commutative diagram across the three different neural network units.

Following our strategy to cover broad ranges of possibilities with respect to variation potentials (vertical, horizontal, and cross variations) The following scheme was arranged:

- Three different fractions of F_A (i.e. 0.4, 0.7 and 1) are considered.
- The input to ABL is in form of $[0.4 * F_A, 0.7 * F_A, F_A]$ while the input to ABT and ABE are in form of $[\frac{1}{0.4*F_A}, \frac{1}{0.7*F_A}, \frac{1}{F_A}]$.

Defining the quantitative measures reflecting relative deviations from:

$$F_A = [1, 1, 1, 1, 1];\ F_B = [1, 1, 1, 1, 1];$$

is done with a combinatory approach using "trace" of matrices as scores. For each run of [0.4, 07, 1] vs [L, T, E] (i.e. 9 pairs) two groups of calculation is done:

1. Changes in F_B with respect to F_A, and the F_{AB}
2. Changes in F_A with respect to F_B, and F_{AB}.

It is important to notice that "changes in F_A and F_B" exactly mean their deviations from identity vectors $F_A = [1, 1, 1, 1, 1];\ F_B = [1, 1, 1, 1, 1]$.

Another important issue is that the neural network units are trained with the input set of F_{AB} where $F_{AB} = F_A + F_B$ however when we feed the input to the neural network to monitor its changes, our input set is actually $F_{AB} = F_A$ because we do not have information about $F_{B.}$ That said, the NN's behavior is unaffected and they will "act" as they are receiving some sort of F_{AB} that is constructed as $F_{AB} = F_A + F_B$. This subtle issue enables us to observe the effects of a specific F_A (i.e. an inquiry) on F_B and quantify its relation with respect to all or nearly all elements or aspects of the process.

As for the quantification method and scoring system the following matrix operation is established. All matrices have the dimension of $[\quad]_{5*5}$.

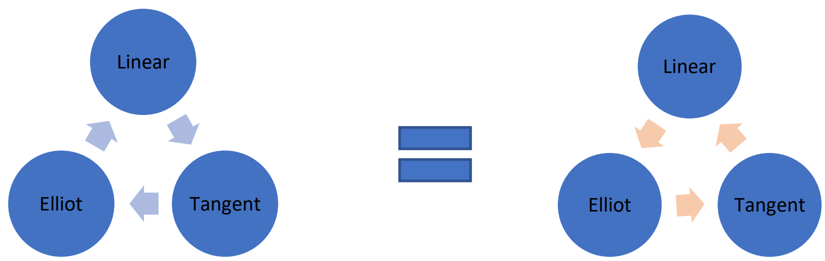
- $[B * A^T]$, $[dB * A^T], \left[dB * {A_{in}}^T\right]$, (reflecting the changes of B to A) (67)
- $[A * B^T]$, $[dA * B^T], [A_{in} * dA^T]$, (reflecting the changes of A to B) (68)

When the trace of the three matrices are summed up in each group, they yield a Figure 50 scores such as below.

		L	T	E
B to A	0.4	1.05	3.94	3.30
	0.7	1.67	3.64	3.00
	1	2.14	3.52	2.90
A to B	0.4	1.12	-1.28	3.43
	0.7	1.71	-1.12	2.94
	1	2.10	-1.04	2.70

Figure 50. Scores calculated from traces of matrices defined by Equations 67&68

Conceptually this figure summarizes the changes along *4 axioms: path, scale, AtB and BtA.* Each score in table 6 is the sum of traces for three matrices ($\sum_{3\,[\quad]_{5*5}} trace$) in aforementioned groups, it can be seen that the information in table 6 is obtained from traces of 54 matrices $[\quad]_{5*5}$ (i.e. 3 scale* 3 path* 6 measure = 54 matrices that overally consist of 1350 numerical value, each probably capturing an infinitesmall relation between A&B) corrosponding to a single inquiry. An important feature of this set up is that forms a commutative diagram between linear, hyperbolic tangent, and elliot modules as illustrated in the next Page (Fig. 51).



		(L-T)	(E-L)	(E-T)	(L-T)+(E-L)
	0.4	-2.89	2.25	-0.63	-0.63
B to A	0.7	-1.97	1.33	-0.63	-0.63
	1	-1.38	0.76	-0.61	-0.61
	0.4	2.40	2.31	4.71	4.71
A t B	0.7	2.83	1.23	4.06	4.06
	1	3.14	0.60	3.74	3.74

Figure 51. Elliot modules as illustrated

An important and inherent aspect of self-ruling functional analyses is to select a point where achievement of entanglement quantification can be fairly speculated. The other important consideration in these type of analyses is to decide the "form" and "dimension" (i.e single scalar, vector, involution, etc.) of the mathematical object ready for final integration. Here, since we are dealing with a convoluted object, it does not seem economical to perform rigorous structural analysis of each case, therefore, we base our judgement on the "general principle" of equilibrium for stationary objects. To be stationary all of the forces on the object must be in equilibrium, the resultant force and moment vectors must both be zero. From structural point of view, the number of equilibrium conditions required to solve different force systems can be summarized as following:

Type of force system	two dimensional	Three-dimensional
General	3	6
Coplanar	3	3
Concurrent	2	3
Parallel	2	3
Coplanar, parallel	2	2
Coplanar, concurrent	2	2
Collinear	1	1

Figure 52. Number of equilibrium conditions required to solve different force systems (Civil engineering reference manual 41-7)

Given our aim to solve a general system, 6 equilibrium condition is needed. By considering all the 4 axioms, the summation shown in the last column in Figure 53 is chosen as the 6 equilibrium conditions.

		L	T	E	Sum
	0.4	1.05	3.94	3.30	8.29
B to A	0.7	1.67	3.64	3.00	8.31
	1	2.14	3.52	2.90	8.56
	0.4	1.12	-1.28	3.43	3.26
A to B	0.7	1.71	-1.12	2.94	3.54
	1	2.10	-1.04	2.70	3.75

Figure 53. Calculation of sum of rows to get 6 static condition

It can be seen that equilibrium conditions for changes of B with respect to A (i.e. the first 3 values) are close to the universal gas constant (R=8.314). This could be partially attributed to involvement of R in construction of the composite variable:

$$N_j = \frac{1}{2}\left(\frac{L_j * V_j}{T_j S_j * lnln\,(R_{j\_max} - R_j)} + 12\right) * \frac{2.3 * R}{F},$$

where T_j was the temperature in ℃, R=the universal gas constant, and F was the Faraday constant, however we are neither to claim that is necessarily the case nor think its correctness or incorrectness significantly reinforce or invalidate our course of actions. A similar argument can be applied with regard to closeness of equilibrium measures for changes of A with respect to B (i.e. the second 3 values) to the mathematical constant π.

That said, π & R can be used to quantify the measure of dispersion around “two” fixed values to normalize the “bimodal” distribution at hand. (In some sense, it is πR-reviewed!).

By normalization of the first 3 values with respect to R=8.314 and the second 3 scores with respect to $\pi = 3.14$, we get the numerical values with the following scale:

	1	2
1	[0.9974;0.9994;1.0290]	[1.0383;1.1259;1.1947]
2	[1.0047;0.9943;1.0162]	[1.0212;1.0961;1.1725]
3	[1.0143;1.0044;1.0274]	[1.0481;1.1511;1.2651]
4	[1.0175;1.0100;1.0354]	[1.0682;1.1863;1.3153]
5	[1.0277;1.0373;1.0773]	[1.1419;1.3093;1.4805]
6	[1.0101;1.0216;1.0608]	[1.0757;1.1913;1.2882]
7	[1.0213;1.0411;1.0887]	[1.0719;1.1846;1.2786]
8	[1.0253;1.0469;1.0928]	[1.1151;1.2655;1.4035]
9	[1.0271;1.0555;1.1081]	[1.0824;1.2032;1.3057]
10	[1.0212;1.0431;1.0875]	[1.0768;1.2099;1.3256]

Figure 54. Static vectors denoted as S_{AB}& S_{BA}

Following this transformation, dispersions around 1, now is supposed to reflect the differential information among different cases. In other words, since “1” is directly associated with 2 fundamental constants π & R , and itself reflects total equity, we may establish the connection between differentials and a fundamental equilibrium state (Equity=1). At this stage, although the non-equilibrium factor in our process has been characterized statically, there remains a fundamental gap: lack of a variable with nature of Rate & Resistance.

Basically, thermodynamics considerations do not establish the rates of chemical or physical processes because resistance is not a thermodynamic variable.

Any proposition to resolve this crucial issue must effectively combine the two seemingly paradoxical concepts of static/equilibrium and dynamics.

Here, we propose a novel technique of analysis based on “abstract physical operation on matrices” with respect to “time”. Let us exemplify the procedure by first examining 4 matrices formed by average multiplication of two static vectors for cases 1-4 as shown below:

$$S_{ci} = \frac{1}{2}(S_{ABi} * S_{BAi}^{T} + S_{BAi} * S_{ABi}^{T}) , \tag{69}$$

$$S_{c1} = \begin{bmatrix} 1.035 & 1.080 & 1.130 \\ 1.080 & 1.125 & 1.176 \\ 1.130 & 1.176 & 1.229 \end{bmatrix}, \qquad S_{c2} = \begin{bmatrix} 1.025 & 1.058 & 1.107 \\ 1.058 & 1.089 & 1.139 \\ 1.107 & 1.139 & 1.191 \end{bmatrix}$$

$$S_{c3} = \begin{bmatrix} 1.063 & 1.110 & 1.180 \\ 1.110 & 1.156 & 1.226 \\ 1.180 & 1.226 & 1.299 \end{bmatrix}, \qquad S_{c4} = \begin{bmatrix} 1.086 & 1.143 & 1.222 \\ 1.143 & 1.198 & 1.278 \\ 1.222 & 1.17 & 1.361 \end{bmatrix}$$

The arrows in the following matrix indicate the direction of increase in the numerical values. This uniform pattern exists in all the 200 cases, and its emergence may be attributed to the overall harmonic tone of our analyses.

$$S_{ci} = \begin{bmatrix} S_{ci}^{11} & S_{ci}^{12} & S_{ci}^{13} \\ S_{ci}^{21} & S_{ci}^{22} & S_{ci}^{23} \\ S_{ci}^{31} & S_{ci}^{32} & S_{ci}^{33} \end{bmatrix}$$

4.5 Institution of a static/dynamic variable & integration

Now, in order to obtain a dynamical property value from a static situation, there is a need for an operation consisting some sort of change without tearing or breaking the static structure of the object. A stretching process to deform the object with condition of sorting the values from lowest to highest value seems to have this particular quality.

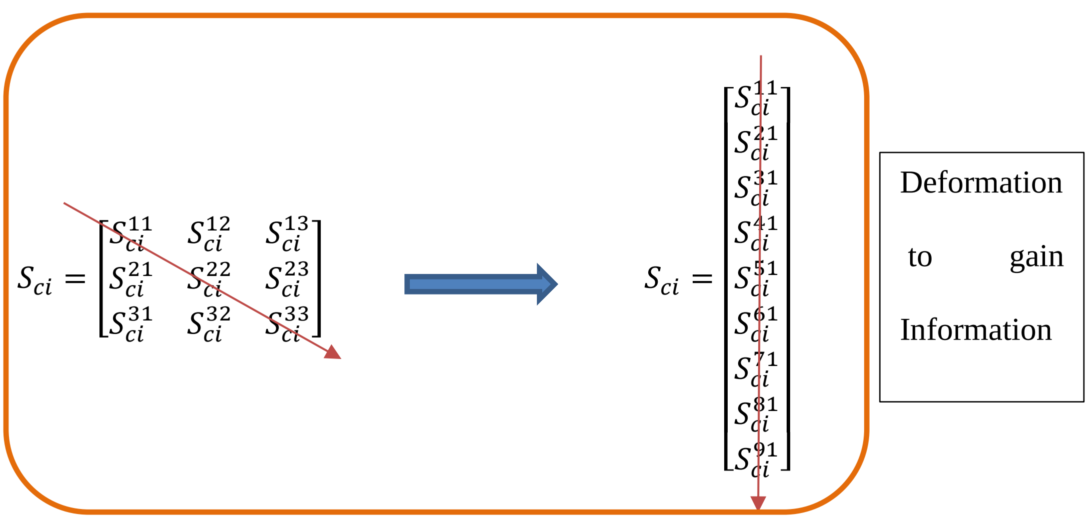


Figure 55. Deformation to gain Information

Therefore, the operation of reshaping the average product of static vectors $[\quad]_{3*3}$ to $[\quad]_{9*1}$, and arranging them from lowest to highest (as if those scores were time series or any other naturally increasing quantity such as entropy) is deemed to represent the act of "change in order", and the duration of time from starting this specific activity till its completion is recorded. This specific computational time varied across different cases. Given the similarity of environmental condition in this test such as the device power, those differences are hypothesized to be related to stiffness of computation for different cases. This hypothesis of "relation to stiffness" is proposed based on (or inspired by) two notions:

1. The analogy between our defined event and the concepts of elasticity and strain for a physical object. (when a material is deformed under a tectonic force)
2. The theory of computational complexity that states:

"In computer science, the computational complexity, or simply complexity of an algorithm is the amount of resources required for running it. The computational complexity of a problem is the minimum of the complexities of all possible algorithms for this problem (including the unknown algorithms).

As the amount of needed resources varies with the input, the complexity is generally expressed as a function n → f(n), where n is the size of the input, and f(n) is either the worst-case complexity, that is the maximum of the amount of resources that are needed for all input of size n, or the average-case complexity, that is average of the amount of resources over all input of size n. *When the nature of the resources is not explicitly given, this is usually the time needed for running the algorithm, and one talks of time complexity*. However, this depends on the computer that is used, and the time is generally expressed as the number of needed elementary operations, which are supposed to take a constant time on a given computer, and to change by a constant factor when one changes of computer”.[85]

Specifically, we utilized an application in the MATLAB environment which allows user to record the duration time of a specific operation. The operation was done for all 200 cases and the time for each case recorded using “timeit” functionality, all the numerical values are given in appendix C.

```
For n=1:200 f = @()
tscollection(reshape(sumrows_LTE{n,3},9,1));tsc=tscollection(reshape
(sumrows_LTE{n,3},9,1));St_En(n,1)=tsc.TimeInfo.Start;
St_En(n,2)=tsc.TimeInfo.End; t1 = timeit(f);  St_En(n,3)=t1; End
```

Figure 56. Function of “timeit”

An important feature that distinguishes this part of analysis from the rest, is its computational irreversibility.

“As was first argued by Rolf Landauer of IBM, in order for a computational process to be physically reversible, it must also be logically reversible. Landauer's principle is the rigorously valid observation that the oblivious erasure of n bits of known information must always incur a cost of nkT ln (2) in thermodynamic entropy. A discrete, deterministic computational process is said to be logically reversible if the transition function that maps old computational states to new ones is a one-to-one function; i.e. the output logical states uniquely determine the input logical states of the computational operation.”[86]

It is important to notice that "inverse functions" can be calculated for trigonometric, geometric transformations, the neural network formulations, etc. However, given a particular "time complexity", the static matrix cannot be back calculated, since the relationship between them is not a bijection (Fig. 57). There is more than one configuration of that matrix with the same computational complexity.

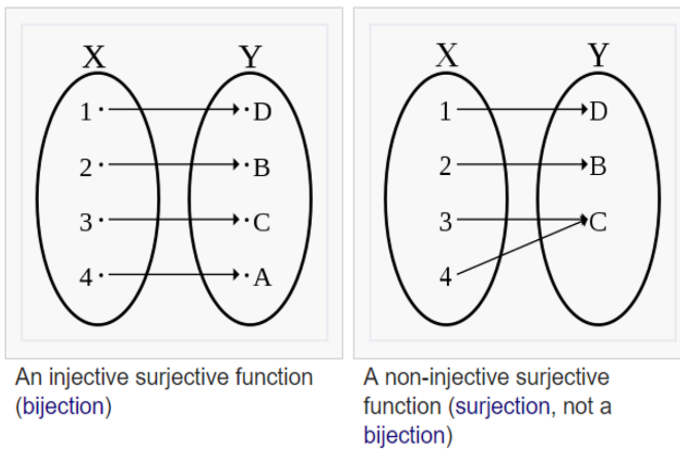


Figure 57. Surjective vs non-surjective function

Nevertheless, attainment of this dynamical/static property value proved to be extremely useful in resulting monotonic trends for important process variables such as carbon conversion efficiency and the yield of product gas (figures 14-17). The specific computational time (sct) marks the finish line of our functional analysis. In the last step, we need to perform an integration with regard to the variables we consider the most critical, and informative regarding the path traveled by our inquiry.

For the choice of integration, given the presence of two focal points in our normalization which was selected due to bimodality of the distribution, a complete "elliptic" integral of the first kind was recognized as the most compatible option. The integration was done through a low pass filter. Specifically, the Ellipap module in MATLAB was employed. This module returns the zeros, poles, and gain of an order "n" elliptic analog lowpass filter prototype, with Rp dB of ripple in the passband, and a stopband Rs dB down from the peak value in the passband (Fig. 38) The zeros and poles are returned in length n column vectors z and p and the gain in scalar k. If n is odd, z is length n - The transfer function in factored zero-pole form returns the poles and zeros, and gain of:

$$H(S) = \frac{Z(S)}{P(S)} = k\,\frac{(s-z_1)(s-z_2)\ldots(s-z_N)}{(s-p_1)(s-p_2)\ldots(s-p_M)} \qquad (70)$$

This module uses ellipke to calculate the "complete elliptic integral" of the first kind and ellipj to calculate Jacobi elliptic. It computes the Jacobi elliptic functions using the method of the arithmetic-geometric mean starting with a *triplet* of numbers. The ellipap module was fed the following three variables which were literally obtained from a *trip* along the *L, E, T* modules:

1. The specific computational time (sct)
2. The summation of 5 Fibbonacci number for an inquiry (A) (i.e. starting point)
3. The absolute temperature (T)

$$a_0 = 1,\ b_0 = \sqrt{1-m},\ c_0 = \sqrt{m}$$

$$a_i = \frac{1}{2}(a_{i-1} + b_{i-1}) \qquad b_i = (a_{i-1}b_{i-1})^{\frac{1}{2}} \qquad c_i = \frac{1}{2}(a_{i-1}b_{i-1})$$

$$\sin(2\emptyset_{n-1} - \emptyset_n) = \frac{c_n}{a_n}\sin(\emptyset_n)$$

$$\mathrm{d}n(u) = \sqrt{1 - m \cdot sn(u)^2} \qquad sn(u) = \sin\ \emptyset_0 \qquad cn(u) = \cos\ \emptyset_0$$

Where the Jacobi elliptic functions are

$$u = \int_0^{\emptyset} \frac{d\theta}{\sqrt{1 - m \cdot sin^2\theta}}$$

$$sn(u) = \sin\ \emptyset \qquad cn(u) = \cos\ \emptyset \qquad \mathrm{d}n(u) = \sqrt{1 - m \cdot sin^2\emptyset}$$

$$[K(m)] = \int_0^1 [(1-t^2)(1-mt^2)]^{\frac{-1}{2}}\, dt$$

Figure 58. The Ellipap module formulation scheme

4.6 Preliminary results

The typical results for poles, zeros and gain are shown below.

	1	2	3	4	5	6	7	8	9	10	11	12	13
1	14x1 comp...	16x1 comp...	16x1 comp...	18x1 comp...	20x1 comp...	18x1 comp...	14x1 comp...	16x1 comp...	14x1 comp...	16x1 comp...	22x1 comp...	16x1 comp...	14x1 comp...
2	15x1 comp...	16x1 comp...	17x1 comp...	18x1 comp...	20x1 comp...	18x1 comp...	15x1 comp...	16x1 comp...	15x1 comp...	16x1 comp...	23x1 comp...	16x1 comp...	15x1 comp...
3	1.4482e-47	3.9811e-53	1.9248e-52	3.9811e-58	1.2589e-60	1.2589e-55	3.1201e-52	1.2589e-55	3.0026e-52	1.2589e-55	3.6470e-53	1.2589e-55	3.4475e-52

Figure 59. Typical results for poles, zeros and gain.

The value of gains displayed in the third row are denoted with Λ. $Ln\ (\frac{\Lambda_S}{\Lambda_G})$ where Λ_G is Geomean of all cases, and Λ_S the value computed for a specific case is a dimensionless number that we call the local minima of algorithm.

The difference between size of pole and zero vectors are either 1 or 0 which we call Squaricty Index (Fig. 60&61).

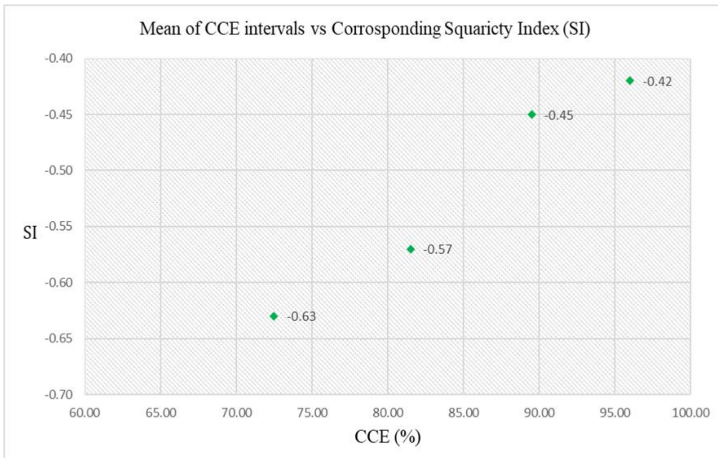


Figure 60. Carbon Conversion Efficiency vs Squarcity Index

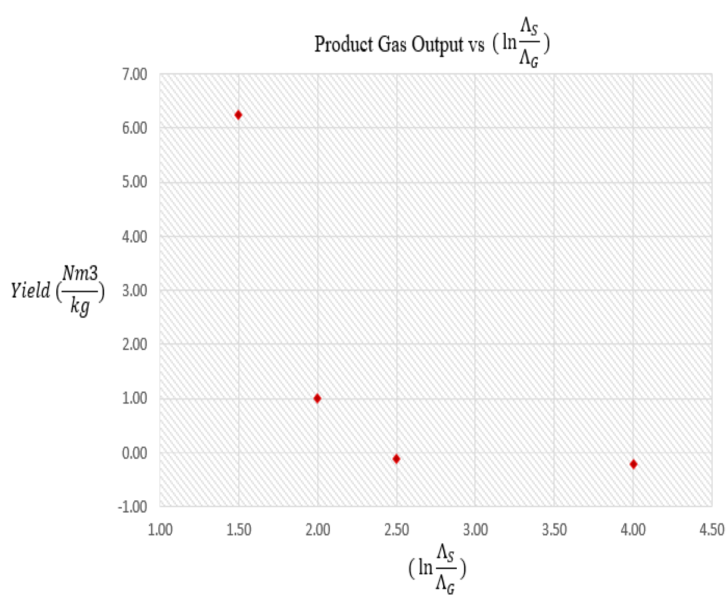


Figure 61. Syngas yield vs local minima of algorithm

For prediction of the product gas composition a neural network module was built utilizing the constructed parameters:

-d
-k
-T
-ER
-SBR

- ER
- ER^a
- ER^b
- b
- $\left|\left(0.55 - 0.7 * max\left\{\frac{T-1000}{T}, \frac{T-1000}{T} + ER\right\}\right)\right|$
- SBR
- $(1 - \log c)$

Where $a = \frac{\sqrt{\left(\frac{6*SBR+10}{\Pi}\right)}}{c}$ $\qquad$ $c = \sqrt{\left(\frac{140*max\left\{\frac{T-1000}{T}, \frac{T-1000}{T} + ER\right\}}{\Pi}\right)}$

$$b = \frac{\Pi * \left(1 + d^2 + \frac{10^3 k^2}{T}\right) * \left(1 - 0.7 * max\left\{\frac{T-1000}{T}, \frac{T-1000}{T} + ER\right\} - \frac{3 * SBR + 5}{\Pi}\right)}{c * 200 * (1 + d^2)}$$

$$y_{H_2} + y_{CH_4} + y_{CO} + y_{CO_2} = 1 \, ,$$

$$cos^2\theta_1 = y_{CO} + y_{CO_2} \, ,$$

$$cos^2\theta_2 = y_{CH_4} + y_{CO_2} \, ,$$

$$cos^2\theta_3 = y_{CH_4} + y_{CO} \, ,$$

$$sin^2\theta_1 = y_{H_2} + y_{CH_4} \, ,$$

$$sin^2\theta_2 = y_{H_2} + y_{CO} \, ,$$

$$sin^2\theta_3 = y_{H_2} + y_{CO_2} \, ,$$

$$y_{H_2} = 1 - \frac{cos^2\theta_1 + cos^2\theta_2 + cos^2\theta_3}{2} \qquad (71)$$

$$y_{CH_4} = 1 - \frac{cos^2\theta_1 + cos^2\theta_2 + cos^2\theta_3}{2} - sin^2\theta_1 \qquad (72)$$

$$y_{CO} = 1 - \frac{cos^2\theta_1 + cos^2\theta_2 + cos^2\theta_3}{2} - sin^2\theta_2 \qquad (73)$$

$$y_{CO_2} = 1 - \frac{cos^2\theta_1 + cos^2\theta_2 + cos^2\theta_3}{2} - sin^2\theta_3 \qquad (74)$$

Where the relationship between $[d, k]$, $[E_1, E_2, S_1, P_1\ ]$, and $[C, H, O, N, S, Cell, Hem, Lig, Vm, Fc, A\ ]$ is as follows:

$$d = \frac{P_1}{3} - \frac{S_1}{3} \tag{75}$$

$$k = \frac{1}{2}\left|\frac{P_1 + S_1}{6} * \frac{2E_1 - E_2}{3}\right| \tag{76}$$

$$E_1 = Ln(\frac{C^{1.45}}{O^{0.17}N^{0.63}S^{0.65}})\ , P_1 = Ln(\frac{FC^{0.52}VM^{0.38}}{A^{0.9}}), E_2 = Ln(\frac{H^{0.88}}{C^{0.31}N^{0.23}S^{0.34}}), S_1 = Ln(\frac{Hem^{0.17}Cell^{0.48}}{Lig^{0.65}}) \quad (77)$$

And $[T, ER, SBR]$ is the gasification operating condition.

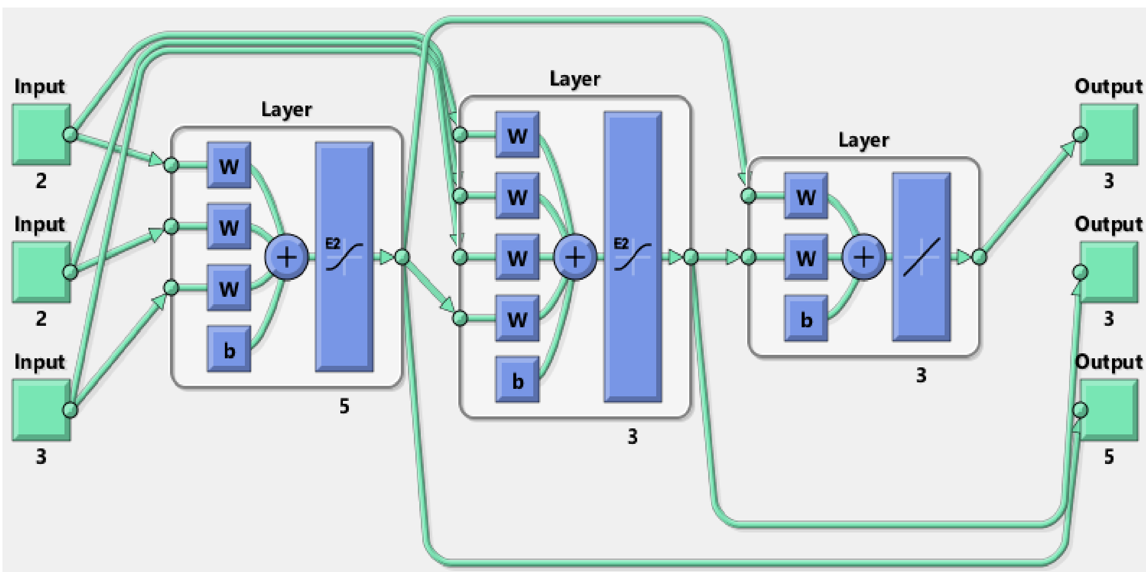


Figure 62. The module developed for prediction of product gas composition

The plots of prediction vs experimental (Overall Error=%8.6) for gas composition is presented in Figures 63-66:

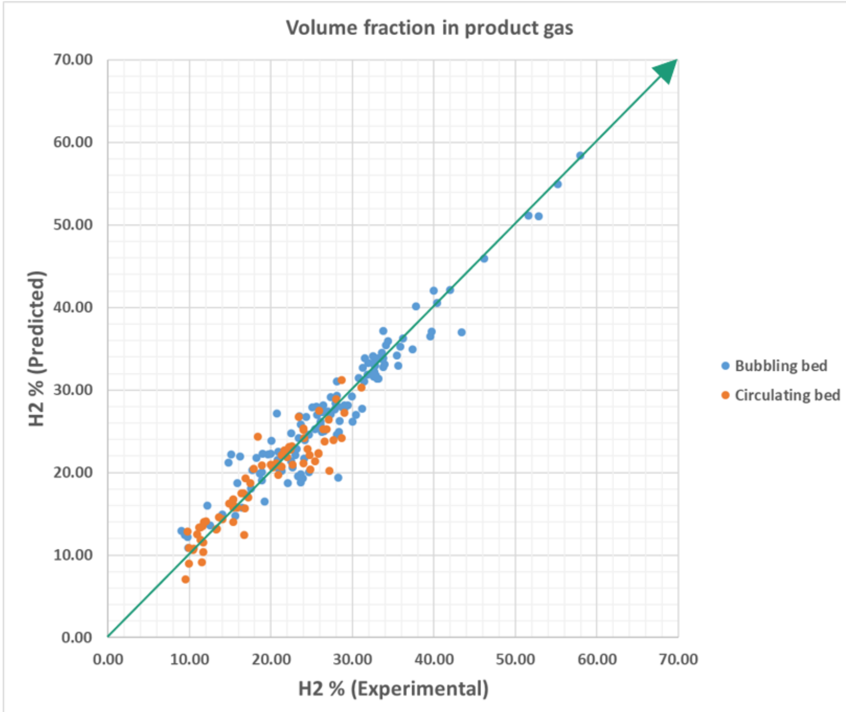


Figure 63. H2 prediction vs experimental values

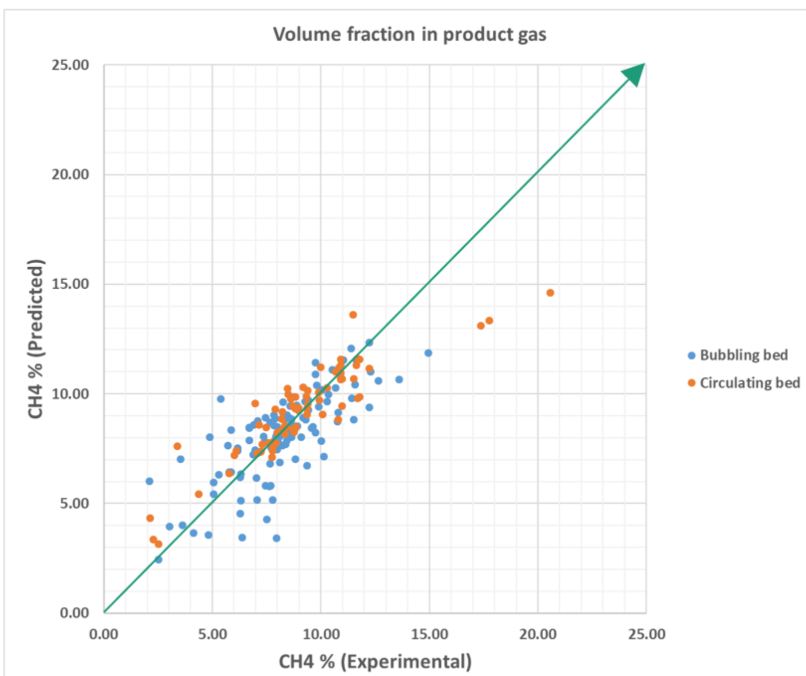


Figure 64. CH4 prediction vs experimental values

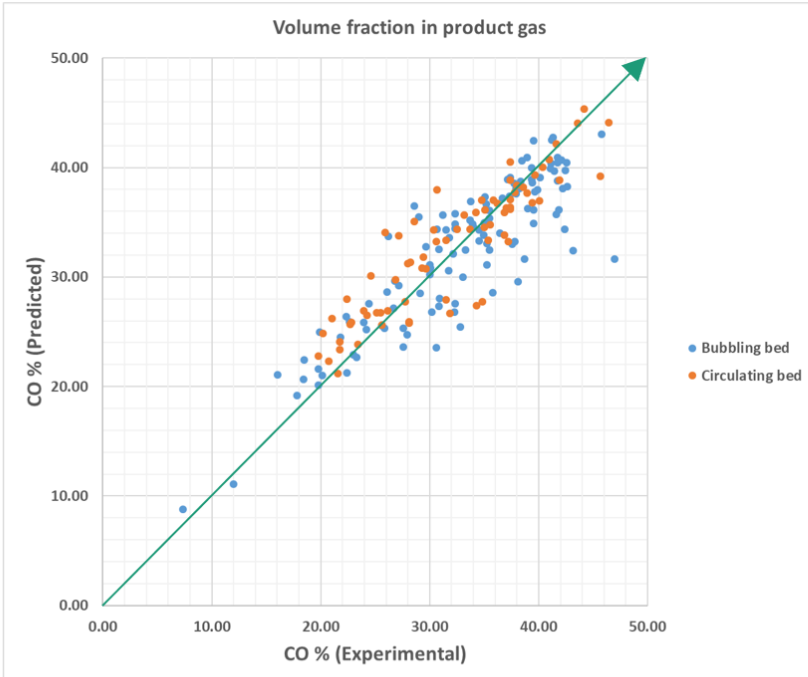


Figure 65. CO prediction vs experimental values

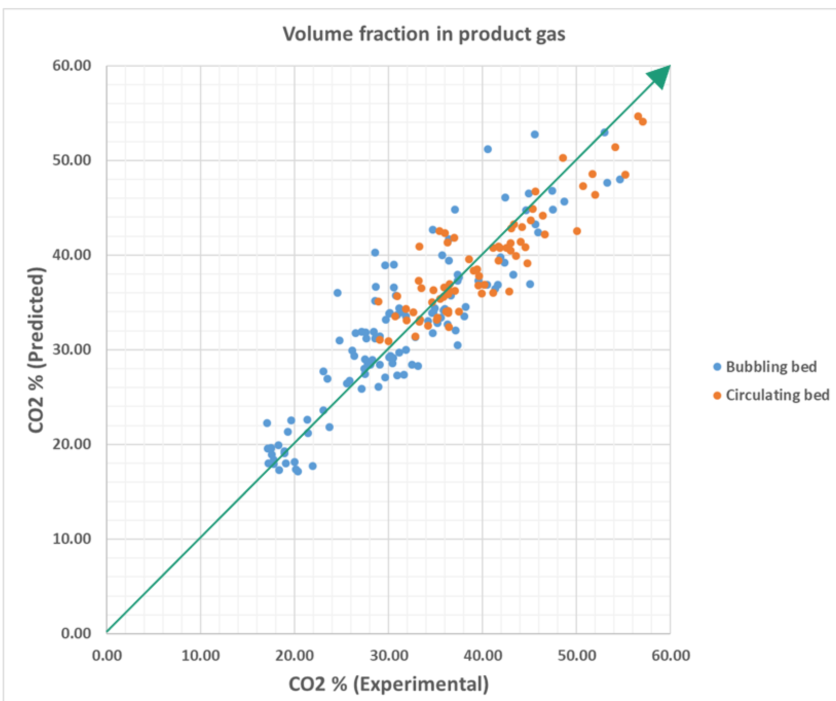


Figure 66. CO2 prediction vs experimental values